

SECURITY IN IOT SYSTEMS

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ABSTRACT

Internet of Things (IoT) has a different meaning to different people. The term has been known for a long time. The use of interconnected devices has become more and more necessary to solve established business problems. It turns out that data acquired from IoT sensors and controllers are valuable enough to have the power to change the overall business models and behaviour of entire industries in a given economy. Therefore, the future of IoT as a cutting-edge technology depends on the security capabilities that IoT systems can provide to its users.

In order to be able to protect an IoT system, we need to know what it represents - what architecture is used, how it works, what kind of components and parts it has, protocols used, main application areas, and so on. Knowing its dependencies, strengths, and weaknesses can provide us a complete and accurate idea of the types of attack vectors to which an IoT systems are vulnerable.

The article presents the vulnerabilities that are typical for different types of IoT systems, the results the attackers strive for, the sequence of stages during the attack process, the potential benefits the attackers would have and the consequences that would have occurred if an attack was successful. Particular attention is paid to the capabilities of protecting a developed IoT monitoring system from such attacks and the ways in which these capabilities can be significantly improved.

Keywords: Internet of Things (IoT), security, attack surface, smart agriculture, beekeeping.

INTRODUCTION

The ability of physical devices to create or connect to a network and exchange data is called Internet of Things (IoT). Each device can be uniquely identified. So the concept of IoT is pretty simple actually. We can find IoT devices almost everywhere:

- HVAC which stands for Heating, Ventilation, Air Conditioning (one of the soundest IoT attacks was done through HVAC);
- Energy sector - windmills, generators, meters, turbines, fuel cells, drills, oil rigs, etc.;
- Consumer and home sector – home appliances, TV, game systems, alarms, locks, lightning and more to come;
- Industrial sector – conveyors, assembly, packing, pumps, valves, tanks, mining facilities, etc.
- Transportation – vehicles, aircrafts, signage, tolls, lights, ships and so on.
- Retail sector – tags, POS, registers, signage, and vending machines.

- Security and Public Safety sector – ambulances, military vehicles, environment, monitoring, etc.
- IT and Network sector - switches, routers, storage, phone systems, servers and more;
- Health care – implants, surgical equipment, monitors, pumps, MRI, tablets;
- Agricultural sector - weather stations, sprinklers controllers, crop management devices, agriculture sensors (collar tags), agricultural drones, sensors in the greenhouse and beehives, etc.

IoT devices can be found in much more places than we probably think, and this makes the requirements to IoT security so important to every one of us, not just the people that are using them.

THE ARCHITECTURE OF IOT

The architecture of an IoT system consists of several different layers [1].

- *Edge technology layer* includes IoT devices, all hardware components such as sensors, readers, RFID tags, actuators, etc.
- *Access gateway layer* – this layer is used to make the connection between the devices and cloud.
- *Internet layer*.
- *Middleware layer* – a bidirectional communication channel including access gateway, internet and backend server or/and storage. The main tasks of this layer are data management, data analysis, aggregation, filtering devices, device discovery, and access control.
- *Application layer* – responsible for making sure of service delivery to users in different areas and places.

Knowing the architectural features of a system allows us to better understand how it is functioning, its strengths and weaknesses. Most IoT systems are designed to collect data from different devices. The collected data goes through the gateway and arrives in cloud server or data storage, where data is stored and analysed. If necessary, the results can be visualized in users' applications.

IOT COMMUNICATION – PROTOCOLS AND TECHNOLOGIES

A protocol is a set of rules to be followed. The communication protocol is the phrase used to describe a set of rules that communication equipment complies with when sending data to other devices. If devices that are trying to exchange data to each other use different set of rules, then the devices will not understand the exchanged data. IoT systems make use of protocols to enable communication between devices themselves and the world [2]. The protocols for IoT systems are mainly divided into four categories – short-range communication, medium-range communication, long-range communication, and wired communication.

Short-range communication technology has low latency and it is also highly reliable. It is secure enough and supports better interoperability. Because of the short range that it utilizes it receives very low levels of interference, even in extreme weather conditions. This is the reason to be widely used for communication between IoT devices. Protocols included in this category are Z-Wave, ZigBee, Wi-Fi, Wi-Fi direct, RFID, Bluetooth

smart (Bluetooth Low Energy, Bluetooth LE, BLE), Li-Fi (Light-Fidelity), NFC (Near-field communication).

Medium-range communication is the same as the communication done with wireless local area networks (WLAN). For the needs of IoT systems, two new protocols are introduced to maintain medium range connectivity at lower power consumption. Wi-Fi HaLow (IEEE 802.11ah) extends today's Wi-Fi range nearly twice while providing better communications through walls. LTE-Advanced is a mobile standard enabling 4G speed of data transfers using mobile carriers.

Long-range communications, shortly LoRaWAN provides low power, long range communication capabilities. The protocol is built on top of LoRa by LoRa Alliance. It allows quick setup of IoT networks anywhere. The networks can be both public and private. Mobile data communication protocols are designed to provide high quality data, but their usage is costly, and they consume lots of power.

Wired communication is the term used when data is being transferred over a wire-based communication technology. Ethernet is a link layer protocol in the TCP/IP stack. Multimedia over Coax Alliance (MoCA) technology uses existing coax cable to convert internet signals to run across that cables.

OPERATING SYSTEMS FOR IOT DEVICES

The smart devices are evolving. This comes with a price – device become more complex than ever. The increasing complexity makes operating on a binary level quite hard, so operating systems (OS) come quite handy [3].

There are several OS for IoT:

- ARM-mbed is typically used in wearable devices.
- Zephyr is mostly used in devices with low resources and low power.
- Ubuntu Core is widely adopted in drones, gateways, robots, etc.
- Apache Mynewt mostly runs on devices that utilize BLE protocol.
- RIOT is free, open source and very friendly.
- RealSense can be found in a lot of different sensors and cameras.
- Nucleus RTOS (real time operating system) feature storage and DB management, USB, multimedia and basic GUI support. Seen in industrial devices, aerospace.
- Brillo is based on Android and used in embedded devices.
- Contiki is used in low power wireless devices like street lightning, sound systems, and monitoring systems.
- Integrity RTOS is designed for multi-core computer. Widely used in military.

THE CHALLENGES IOT PRESENTS

Researchers focus mainly on the communication aspects of the Internet of Things. Most of the research put their attention to the areas of real-world performance; little attention is paid to the integration of IoT devices with software components as resources in various information processes and security. There is currently no standard methodology for modelling and dealing with real complex scenarios using IoT. The unresolved issues that are available in IoT systems largely predetermine the major threats to which IoT

systems are placed. The challenges in concerning the security aspects in the design and use of IoT systems are summarised in Table 1. These challenges open doors to a breakthrough in system security. In the Table 2 possible IoT threats are shown.

Table 1. The challenges that IoT presents

The Challenges	Description
Interoperability standards	There are so many devices, so it is hard to follow standards. Some manufacturers are NOT concerned about security at all.
Clear text and open ports	No encryption implemented for transmitting data.
No basic security or privacy	No support for device authentication or authorization
Support for firmware and OS Updates	It is still difficult to update firmware of already embedded devices and it is mandatory that firmware should be updatable from trusted sources only.
Storage issues	Devices are so small so there is no enough space; there are issues about data management and persistence.
Default, weak credentials	Even hardcoded and easy to break.
Theft and tampering	Hard to physically protect the access to remote devices
Insecure web interfaces	Lots of devices have a web interface built-in them.
Buffer overflow issues	Buffer overflow or some other injection type attacks possible
Development issues	Hard to find professionals with skills relevant to IoT.
Vendor support	Device is only good if vendor is known and supports it.
Regulatory/Rights issues	This is more of “what can happen” issue because of all interconnectivity of IoT devices. If some issue happens, then there might be no law defined to handle it in a proper way.

Table 2. Types of IoT threats

Type of IoT Threats	Description
Insecure network services	Buffer overflow or DDoS attacks, etc.
Authorization/Authentication	Poor or no passwords, etc.
Insecure web interface	Using HTTP. CORS, XSS, CSRF, etc. vulnerabilities
Absence of encryption	Especially during data transmission.
Physical security	Most devices are deployed on remote sites and can be accessed physically and tampered.

Privacy Issues	Some devices are gathering lots of private and confidential data.
Insecure mobile apps	Lots of app do not require passwords to launch an application, there is no 2FA (Two Factor Authentication).
Cloud interface	Insecure interfaces or stored in browser passwords, etc.
Configurability	Is there logging setup, etc. Configuration Manager provides remote control, patch management, software distribution, operating system deployment, network access protection and hardware and software inventory.
Firmware/Software	The ability to update firmware or software as time changes and need is present is imperative.

THE ATTACK SURFACE

The attack surface describes all the different points in which the attacker can penetrate a system in order to change its behaviour and/or steal data. Because of its very nature, IoT has a massive attack surface. Knowing the attack surface well will help us be better prepared for the possible attacks the systems can undergo.

Most of IoT devices suffer from low memory. They have memory just enough to operate. It is quite common that such devices store encryption keys and credential in their memory even in plain text.

By default, all components in an IoT system trust each other. If one is in control by an attacker, the entire system can be easily taken into control also. This type of implicit trust is a huge vulnerability. Also enrolling a new IoT device into an environment (IoT system), the attacker can use this to hook to a given system and try to take control over it. Most devices communicate by using a gateway; if this gateway is compromised the entire system is compromised also.

Physical interfaces like USB ports support plug and play. They can include both user and administrative command line interfaces. It is also often happening that users have administrative rights without being an administrator. This does not prevent privilege escalation scenario when one user can gain admin rights.

Most manufacturers pay more attention to the fact that their devices have more features than security. Features that are firmware dependent potentially have hard-coded credentials which can be easily extracted using firmware dump.

Updating is one of the biggest downfalls that are present. Users rarely make sure that updates are legit but not from untrusting sources. If updates downloads are not encrypted, then this is a great security issue for all that use this type of devices.

Web interfaces related to IoT systems suffer to common security issues that web applications have. There always be a thread attacks like injections, scripting, CSRF (Cross-site Request Forgery), brute force credentials, weak passwords, inability to lock

accounts, etc. Network services allowing IoT devices to talk to each other can be vulnerable to DoS attacks.

THE PROCESS OF IOT HACKING

With the knowledge of the tools that are in possession of the hackers, it is also very important to know what the possible benefits for hackers are, what are mostly hackers interested in. Most often it is all about getting data, ransomware, malicious events that might cause, security breaches or damages or any other undesired effects to a system. Pivot points are also targeted by hackers because they are often not heavily monitored and once in the network, hackers can move on without much risk of being caught. Last, but not least is the prying/spying on monitoring devices like security cameras, microphones, etc.

There are no rules or best practices when talking about how to hack a system. Hacking is thinking outside the box and requires lots of knowledge and creativity. That makes it so difficult to prevent systems from being hacked. Attacks usually undergo in different phases.

Phase 1: Reconnaissance – looking at the target before it is engaged. It can be passive with no direct interaction with the target. Online resources, research papers, websites, documentation available, social engineering, and so on are used during that phase.

Phase 2: Looking for vulnerabilities – when enough information is gathered from reconnaissance with the help of several tools like port or other vulnerability scanners, hackers are trying to gather info about identity systems and other vulnerabilities. Any vulnerability allows creating a new attack vector.

Phase 3: The attacks – when vulnerabilities found, then attacks are deployed against them. There are so many variations of the attacks. The IoT new specific attack is the Rolling Codes widely used in automotive industry (key locks, etc.). It is done with using various tools like RF-Crack, jammers, scanners, re-players, etc.

Phase 4: Achieve access – used paths are via not well secured networks, vulnerable or rewritten OS or applications.

Phase 5: Maintaining access, stay anonymous – maintaining access is just as important as gaining access. So, hackers put a lot of effort to achieve taking full control over (PWN) the system. Cleaning traces by deleting logs allows hiding the successful attack for a longer time. Using the compromised system as a pivot point allow gaining access to other resources, rewriting the firmware with built-in backdoors, unlimited use of the device or system resources. Probably the most important step that is taken when the system is breached is to harden it up to make it harder for other possible attacks from other hackers.

COUNTERMEASURES

The IoT world exposes users at increased risk on a daily basis. Countermeasures must be taken now. It's not just about using the strongest passwords and just using applications that work with optimal encryption. We can do more, we need to learn how to detect malware and ransomware and invest in software that can protect users' data and systems by implementing up to date best practices and using the best tools market provides.

There are some very helpful common-sense practices that are valid not just for IoT systems but also for all systems that use internet and rely on remote access and using them can save all of us lots of troubles. Always change default credentials and never rely on defaults, make sure devices implement lock-outs, use strong authentication, isolate IoT networks from other production and industrial networks if possible, deploy intrusion protection/detection systems, use PKI/End-2-End encryptions and VPNs along with IP whitelisting always when possible, watch out telnet services and port 48101, turn off UPnP (universal plug and play) and automatic discovery. Update not only the gateways but also the devices because lots of breaches can be stopped if latest security patches and updates are installed on time. Restrict physical access to the system and devices – it is said that if attacker has physical access to your device, you do not own this device any more.

Additionally, we must strengthen IoT systems by using storage encryption and gateways must communicate with devices using encryption. The cloud with secure Web UI (using https and encryption) has to be the main place where the data from IoT devices is aggregated and stored for further analysis. Using 2FA, TLS or SSL when possible is highly desirable.

Make sure updates are not coming from unknown sources and how we handle the update process (Chain of Trust Updating). Use remote locking feature when there is no need of accessing the device remotely. Turn off unused services, ports, apps and features. Secure Boot Chain feature protects the device from running a tampered version of the software by verifying its authenticity.

It is also very important to follow manufacturer guidelines to be sure that IoT devices are not exposed to threats.

WHAT WE DO – HOW SMARTBEEHIVES CARES ABOUT THE SECURITY

The concept of the Internet of Things is at the heart of intelligent agriculture and smart bee-keeping plays an important role in it.

The developed innovative IoT system SmartBeeHives [4] takes all necessary measures to protect the data of its users. From the analysis provided in the article on the benefits that hackers are hitting and the functionalities which SmartBeeHives offers to its clients it becomes clear that the targets for the attackers are the data that the system collects from the IoT devices and the communication infrastructure built for the needs of the system. User data can be used for ransomware or other purposes, and through the communication infrastructure, if it is breached, hacker attacks such as DDoS can be launched against other systems. Based on these observations, the vectors of attacks can be clearly distinguished to which SmartBeeHives is vulnerable.

According to The Target Breach Case Study [5], key security details of a system should not be disclosed in public resources. That's why SmartBeeHives do not disclose in detail the concrete measures that are being taken to maintain a high level of security. Only the key elements of the system are marked with special attention.

SmartBeeHives uses secure web interfaces with strong authentication and authorization mechanisms and notification system, secure network services with encryption in communications, minimum personal information is gathered from IoT devices and it is

kept away from them, firmware is coming from the vendor itself with digitally signed updates, improved physical security and much more.

CONCLUSION

The IoT is growing rapidly and the various industries are increasingly relying on the devices connected to the Internet in their business. Research in this area is largely focused on real-time data acquisition, on-site data manipulation, effective data detection [6] and real-time visualization. The availability of data generated by a variety of IoT sources opens up new opportunities for innovative applications in various fields and overall decision-making [7]. But these opportunities should not be compromised by possible breakthroughs in security.

General consideration of security aspects of IoT devices and systems is provided in the paper. The problems concerned to IoT challenges, possible pathways to breakthrough and measures to counter them are described. These issues depicted in the article are reported in the developed IoT-based system – SmartBeeHives. The system is one of possible IoT solution to agricultural problems. But security topics in smart agricultural are common to all kinds of IoT developments and there are common ways of preventing them [8]. Attention should also be paid to the prevention of future unknown potential threats.

REFERENCES

- [1] Varga E., Draskovic D., Mijic D. Scalable Architecture for the Internet of Things. O'Reilly. 2018
- [2] Bhuvaneswari V., Porkodi R. The Internet of Things (IoT) Applications and Communication Enabling Technology Standards: An Overview. International Conference on Intelligent Computing Applications, pp 324 – 329, 2014.
- [3] Stallings W., Operating Systems: Internals and Design Principles. Pearson. 2018
- [4] Dineva K., Atanasova T. Computer System Using Internet of Things for Monitoring of Bee Hives, 17th International Multidisciplinary Scientific GeoConference SGEM 2017, Vienna GREEN Conference Proceedings, Vol. 17, Issue 63, pp 169-176, 2017.
- [5] Dominguez S., The Target Breach – Case Study, Lessons Learned and the Lockheed Martin Intrusion Kill Chain Model, IBM Lab Services, <<http://www.bucks.edu/media/bcccmcdialibrary/con-ed/itacademy/The-Target-Breach.pdf>>, 2016.
- [6] Alexandrov A., Monov V., ZigBee smart sensor system with distributed data processing, Proc. of the 7-th IEEE International Conference Intelligent Systems IS'2014, Vol. 2: Tools, Architectures, Systems, Applications, Warsaw Poland, Advances in Intelligent Systems and Computing, Springer, Vol. 323, pp 259-268, 2014.
- [7] Balabanov, T. Long Short Term Memory in MLP Pair, Proceedings of International Scientific Conference UniTech 2017, Gabrovo, Vol. II, pp 375-379, 2017.
- [8] Blagoev I., A Method for More Reliable Users' Authentication in Internet, Proc. International Scientific Conference, NVU, Veliko Tynovo, Bulgaria, Tom 9, pp. 167-176, 2018.